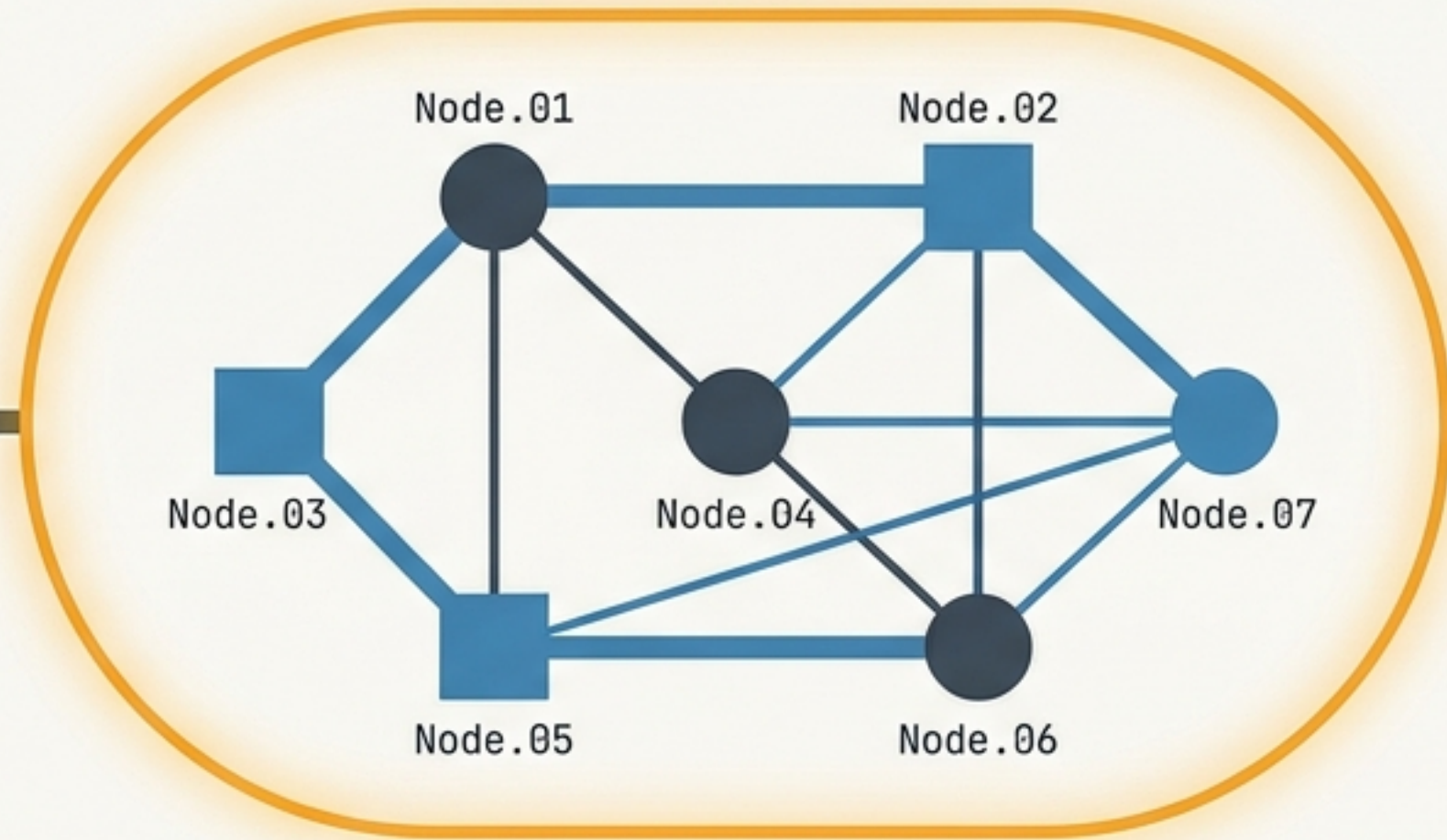


The Context Packet

A Network Primitive
for Distributed AI.

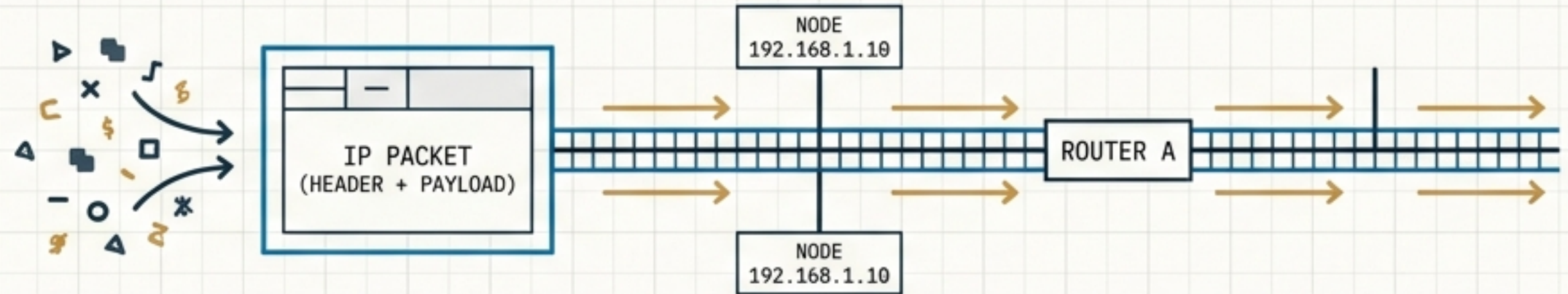


```
ContextPacket<Graph> context_data;
```

THE INTERNET'S LESSON: ABSTRACTING COMPLEXITY

1970s: THE INFORMATION NETWORK

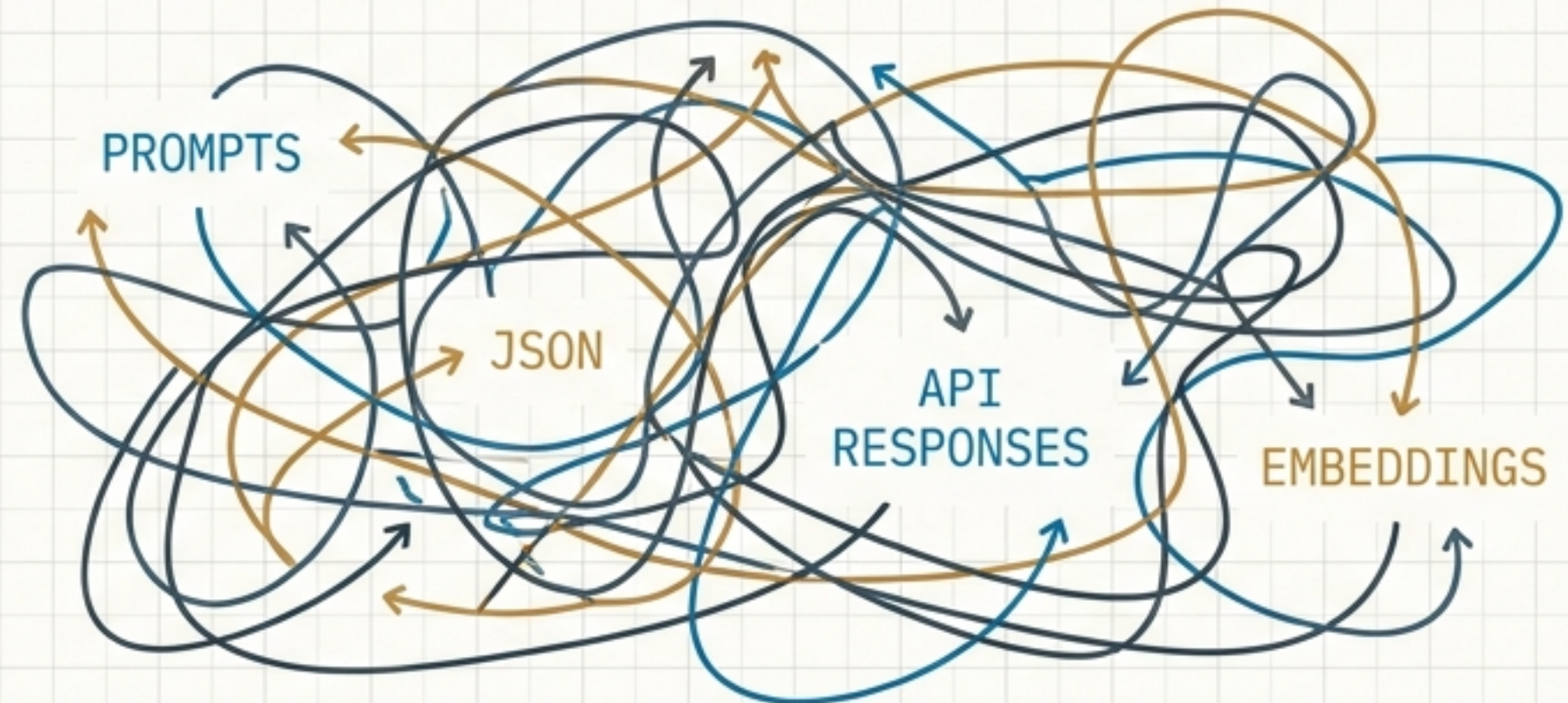
The internet scaled because applications didn't need to understand the network. They just needed bounded, routable packets.



2020s: THE AI NETWORK

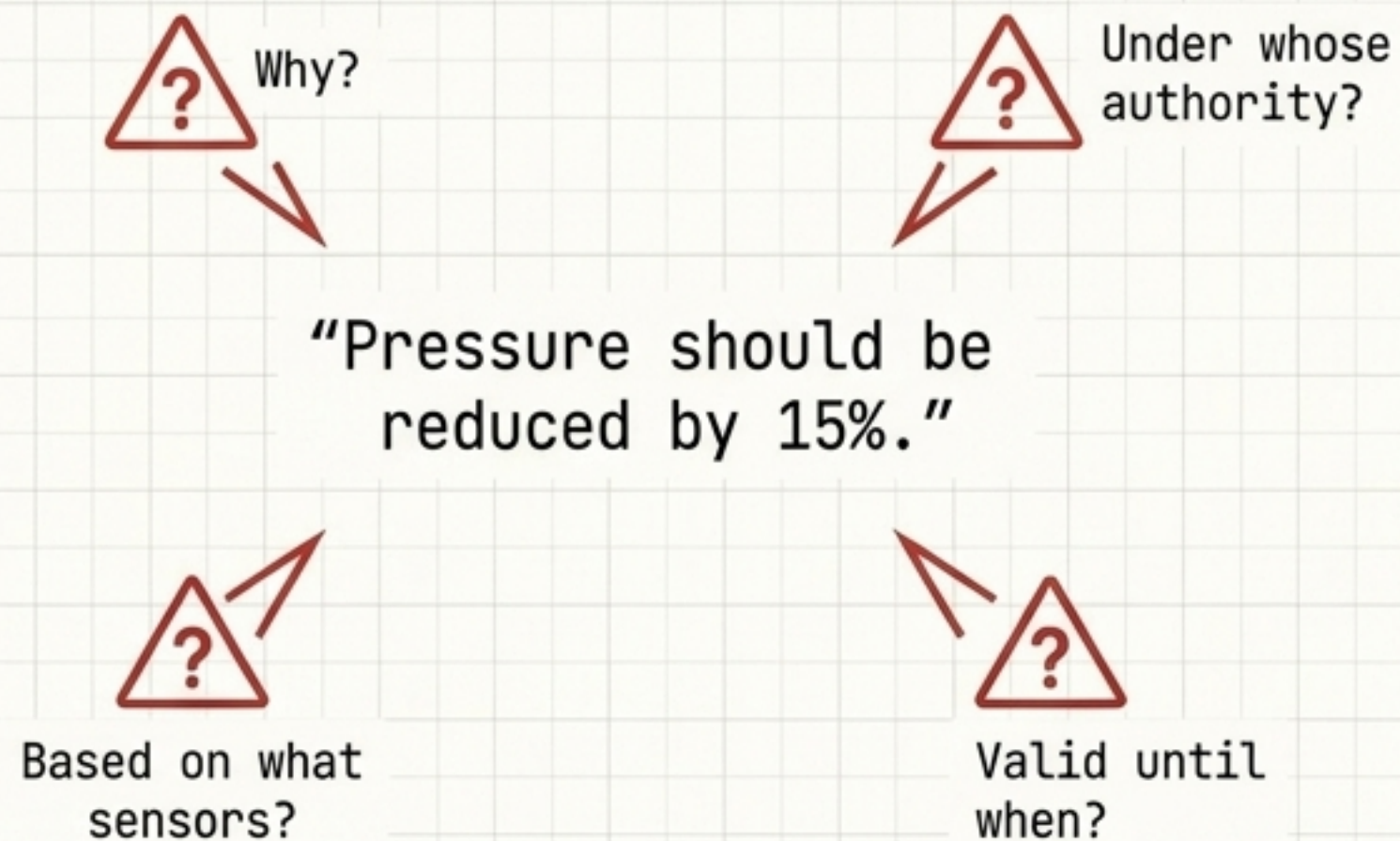
Distributed AI lacks this primitive.

We exchange formats that move information, but fail to move meaning.

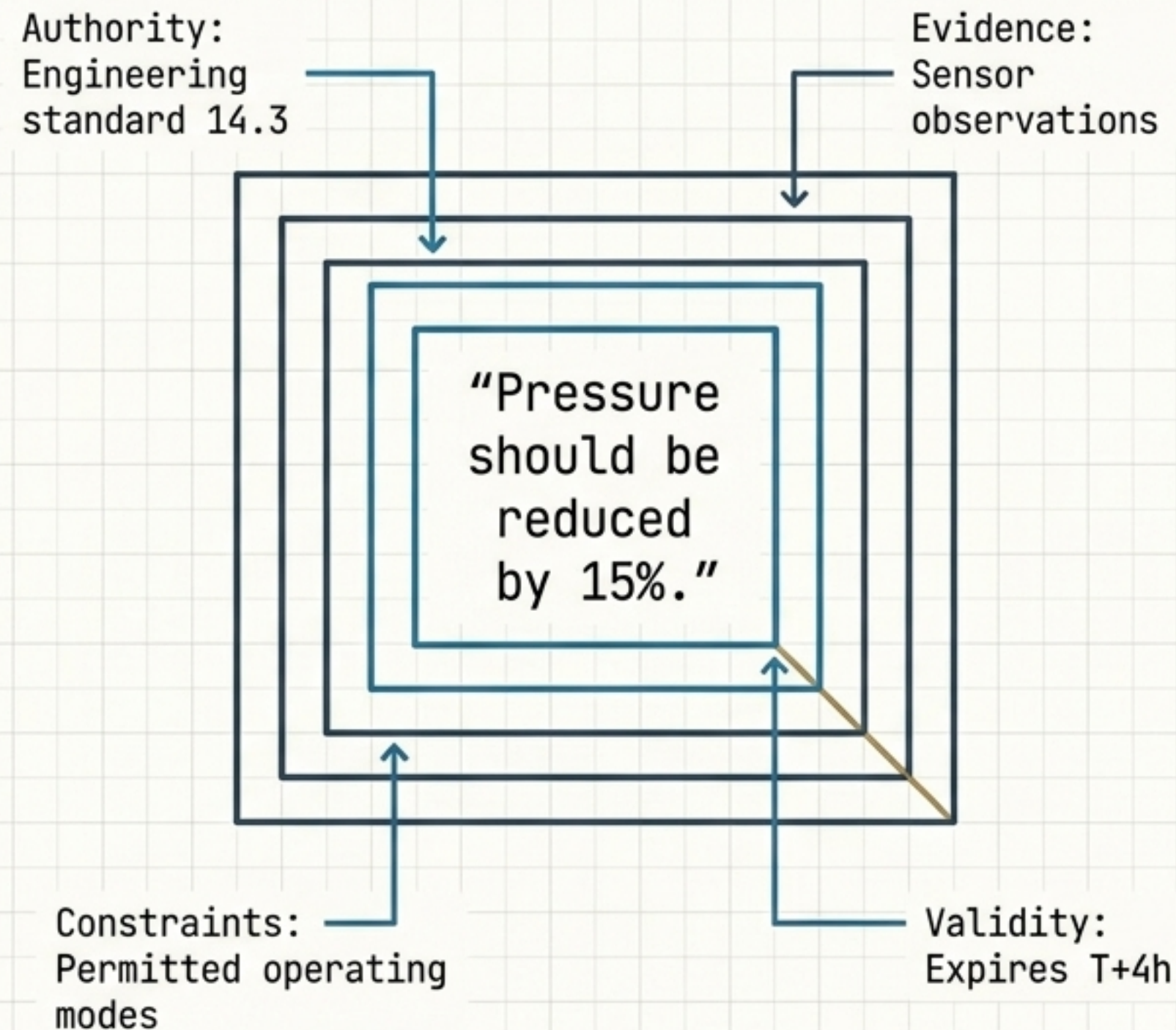


THE CONTEXT GAP: INFORMATION $\neq$ DECISION

NAKED OUTPUT



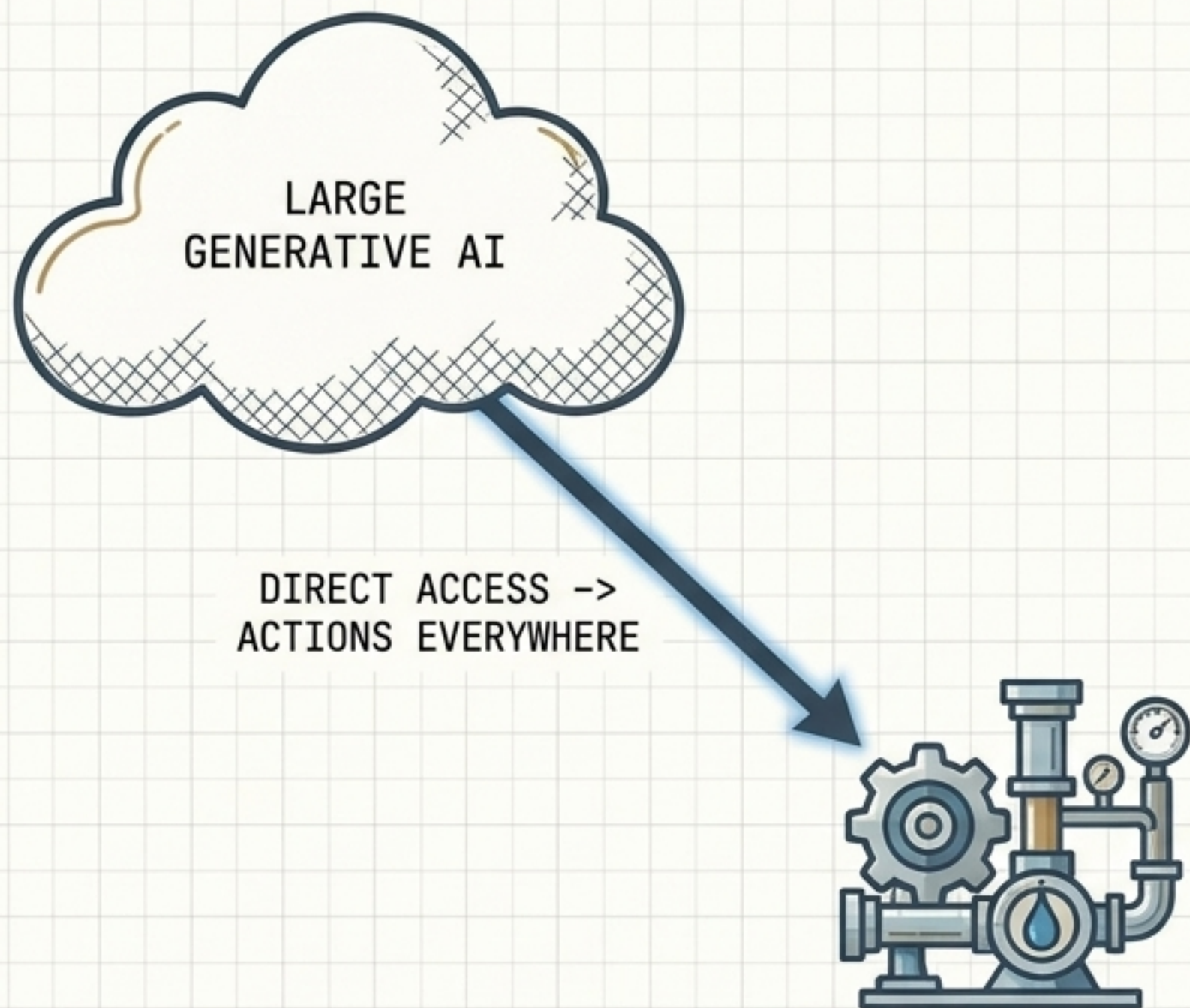
ARMORED CONTEXT



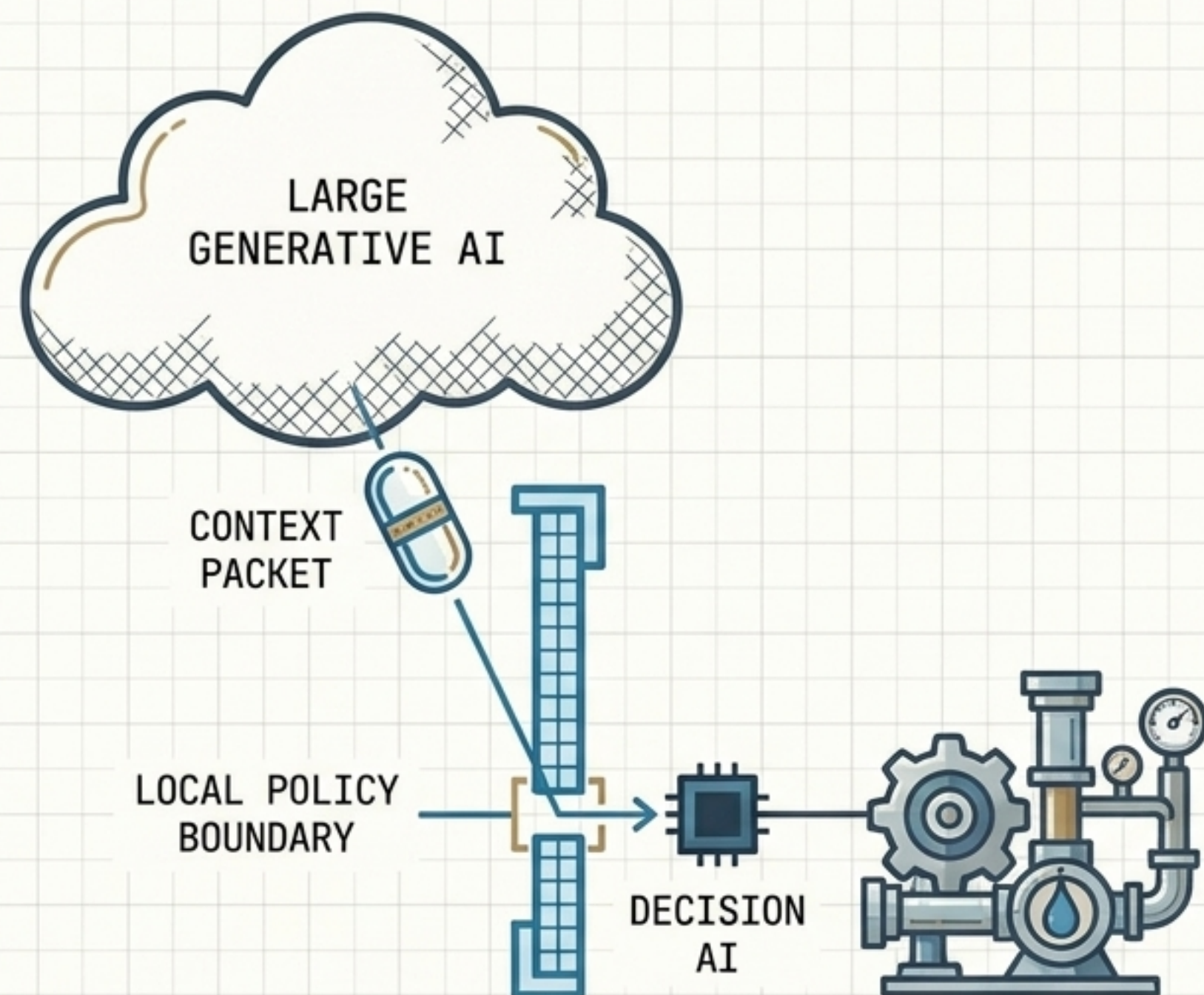
THE DANGER OF UNCONSTRAINED AGENTS

Inserting a structural boundary between generative reasoning and real-world authority.

THE FLAWED PATH

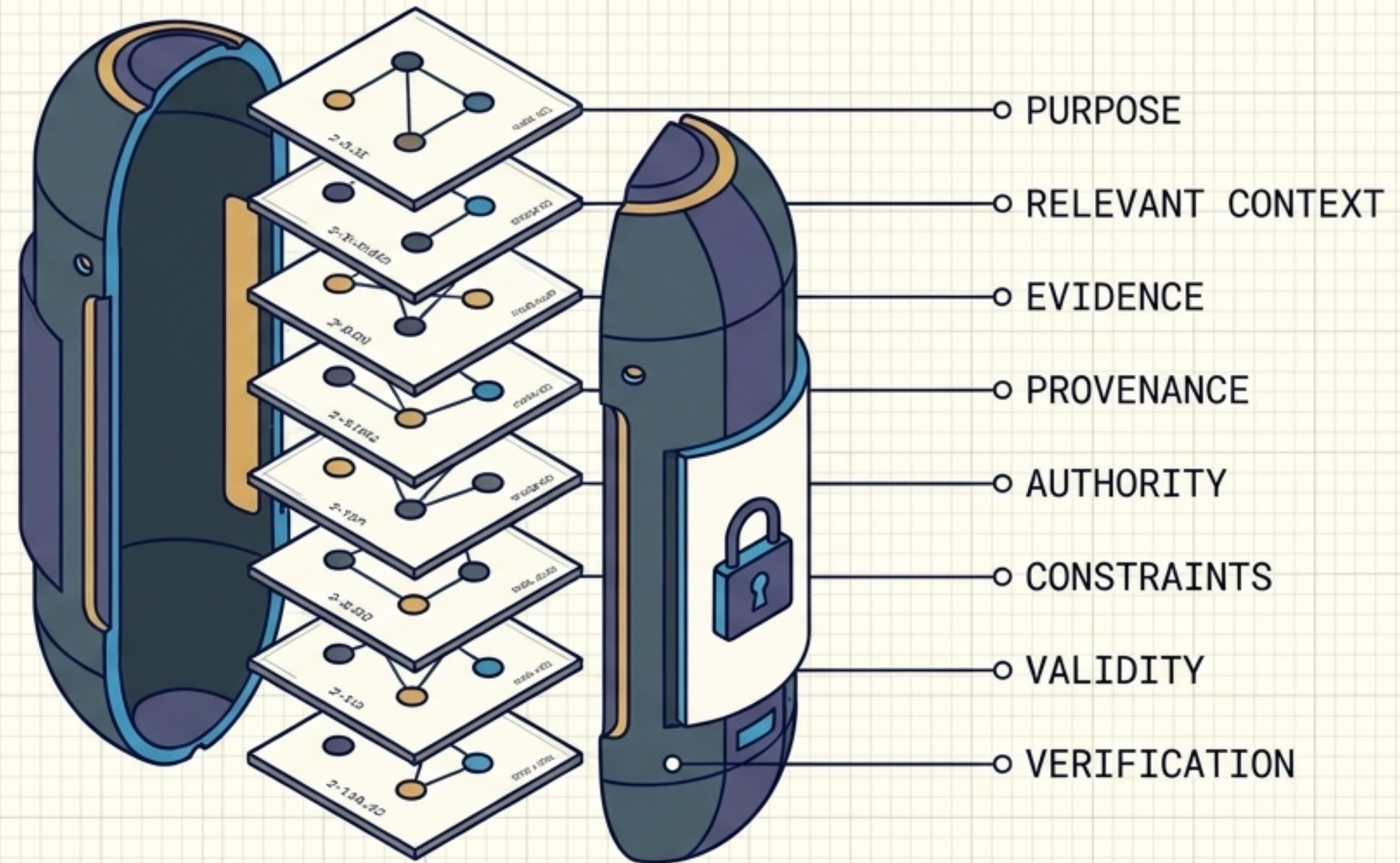


THE SOVEREIGN PATH



ENTER THE CONTEXT PACKET

A bounded, portable, verifiable subgraph containing the minimum context required by an identified actor or system to understand, evaluate, or act upon a specific purpose.

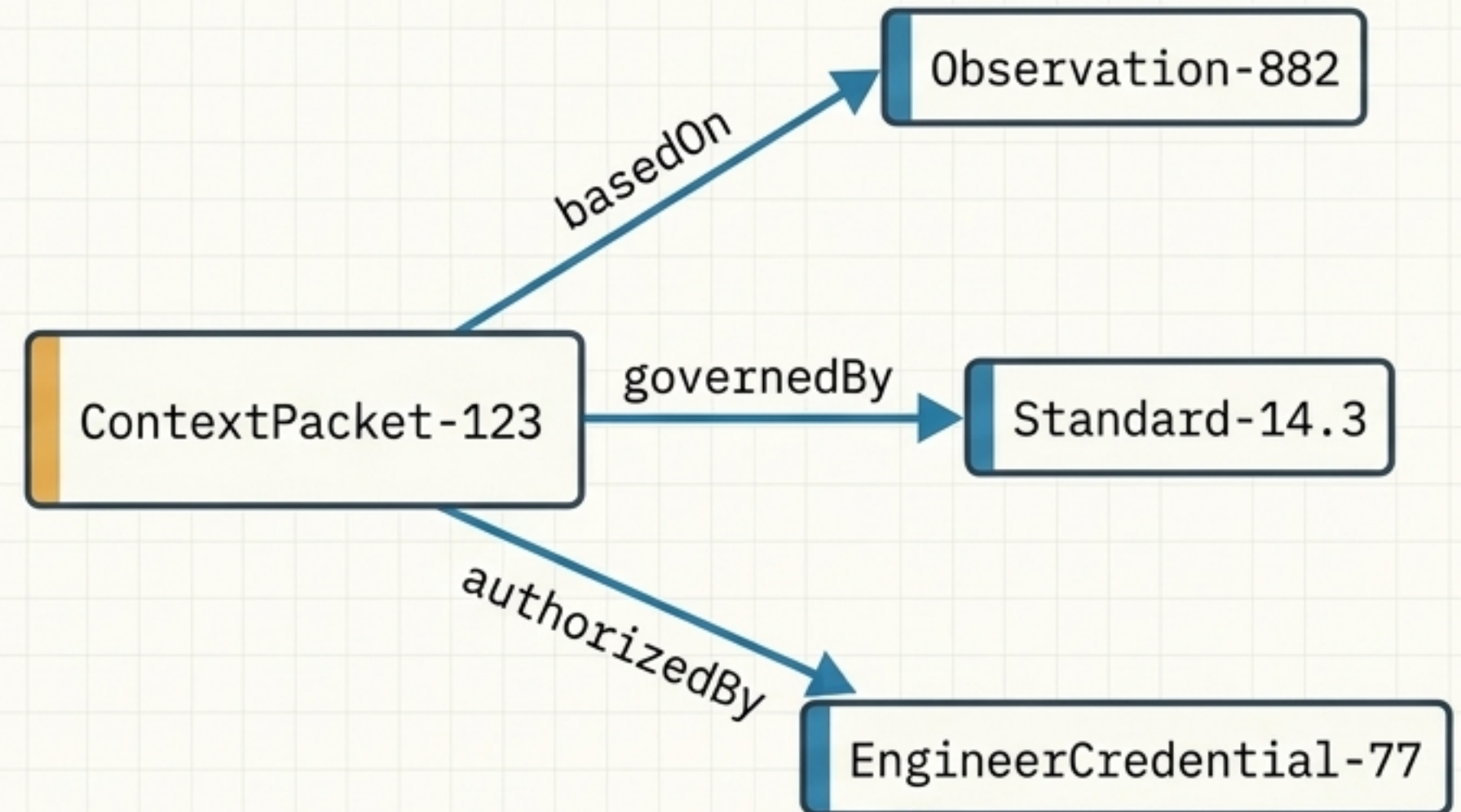


DATA STRUCTURE REALITY: THE PACKET IS A GRAPH

CODE REPRESENTATION

```
{  
  type: context-packet,  
  id: ctx:7a21...,  
  subject: {id: asset:pump-p204}  
}
```

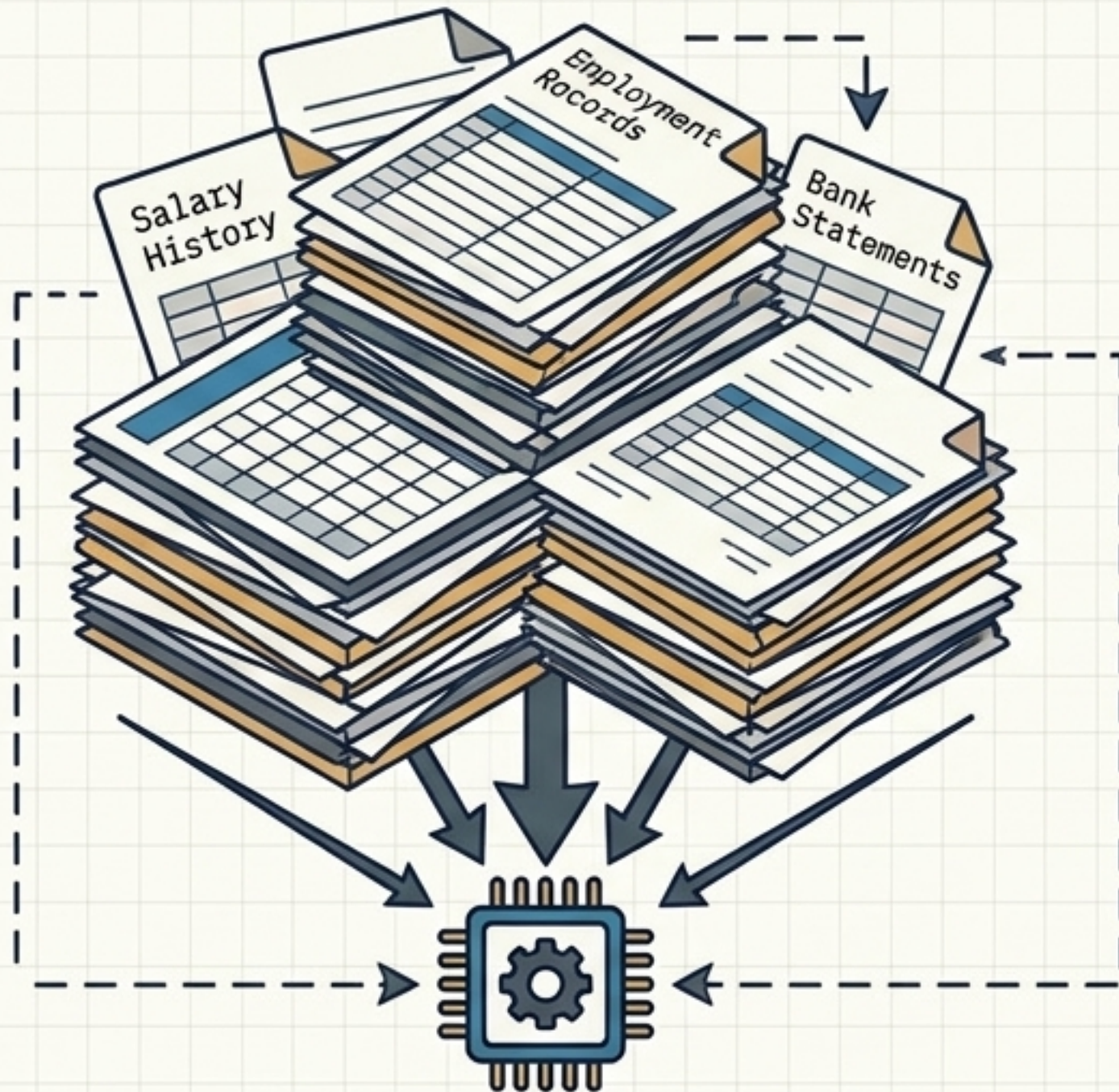
RELATIONAL SUBGRAPH



THE PRINCIPLE OF CONTEXT MINIMIZATION

Abundant intelligence does not imply abundant data sharing. It enables the exact opposite.

DATA MAXIMIZATION



CONTEXT MINIMIZATION



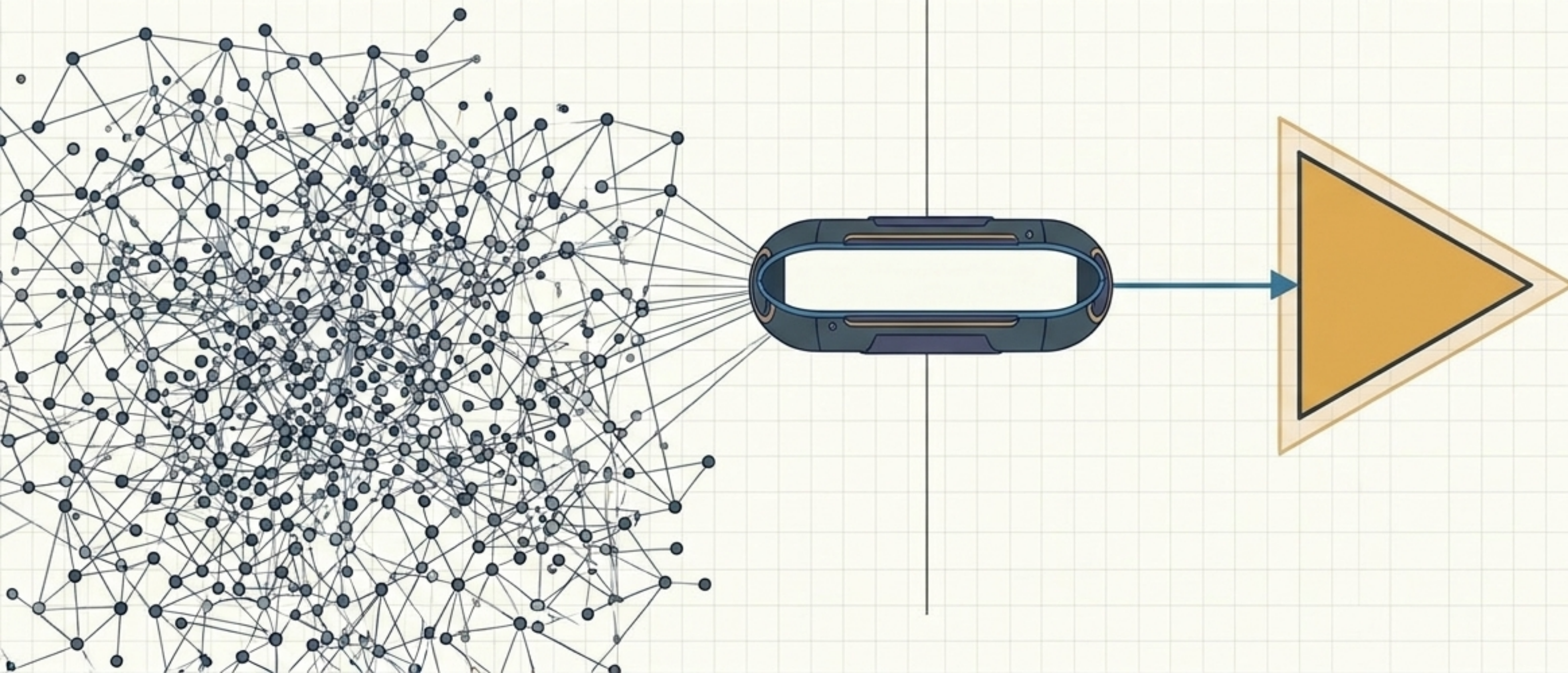
REDEFINING THE AI ARCHETYPES

GENERATIVE AI

Explore, Synthesize, Discover, Hypothesize
Compiles the decision environment

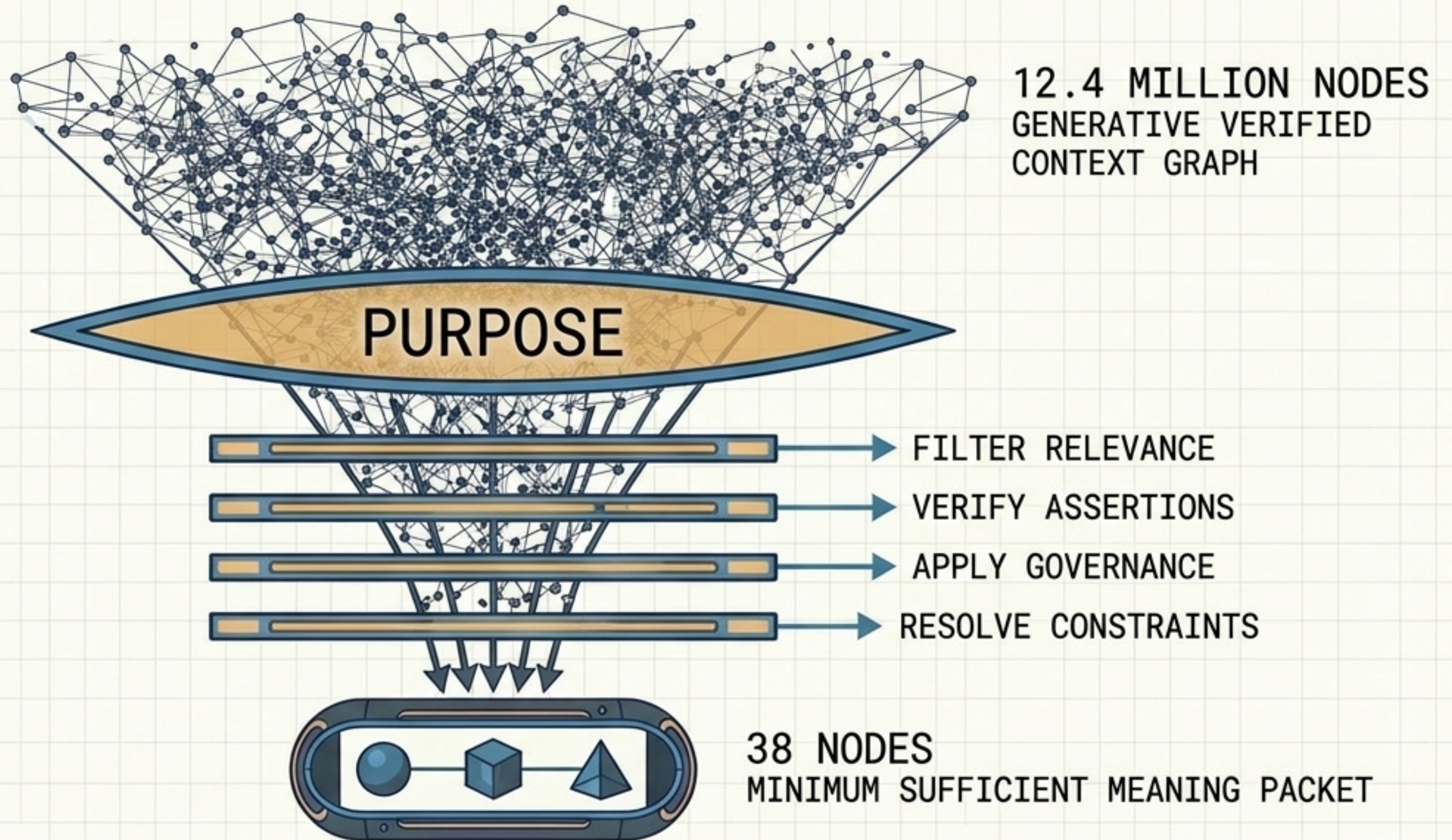
DECISION AI

Evaluate, Constrain, Act, Determine
Executes within the boundary

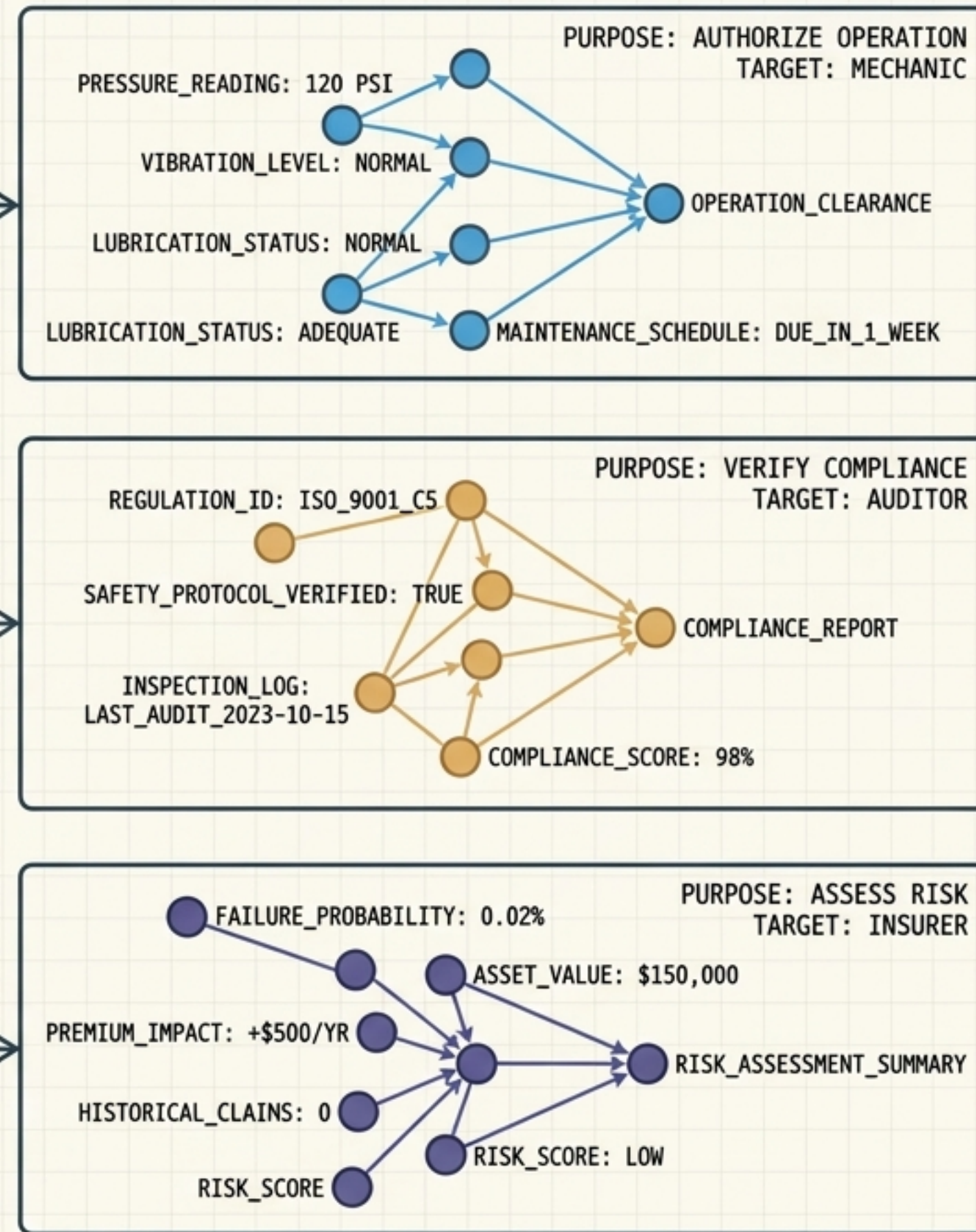


THE CONTEXT COMPILER PIPELINE

From massive global context to minimum sufficient meaning.

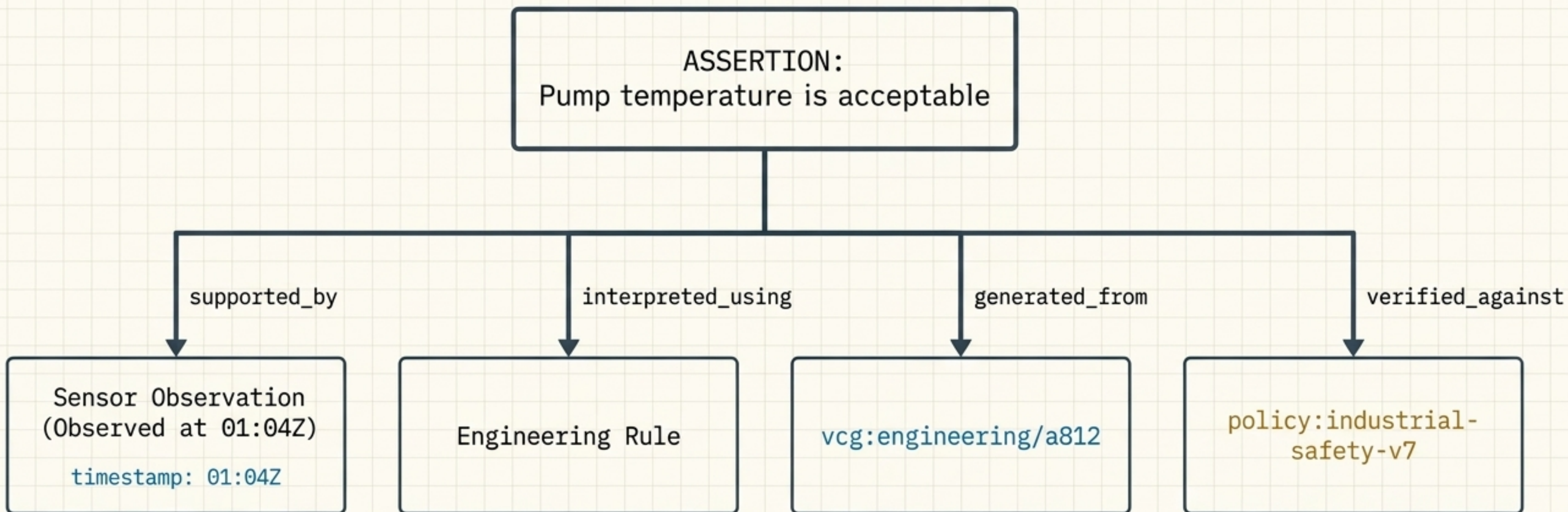


There is no universally complete context.
There is only context sufficient for a purpose.



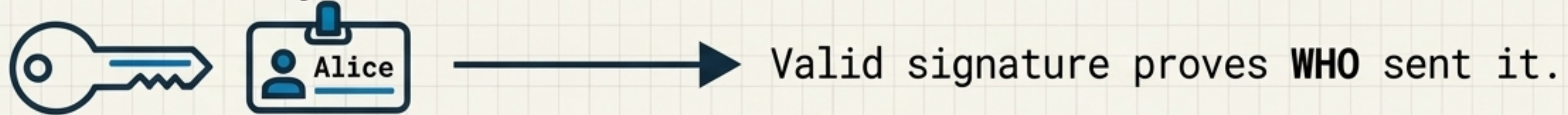
THE ANATOMY OF TRUST

Provenance allows receiving systems to inspect lineage rather than blindly trusting final outputs.

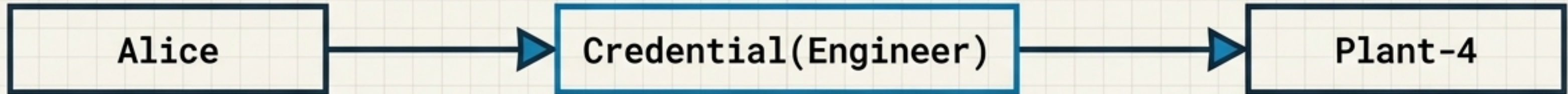


SIGNATURES PROVE ORIGIN, AUTHORITY PROVES RIGHT

Phase 1: The Key

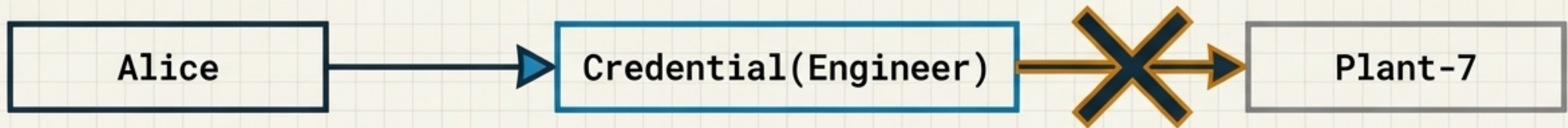


Phase 2: The Blueprint



The graph proves **WHAT** they are authorized to assert about **WHICH** subject.

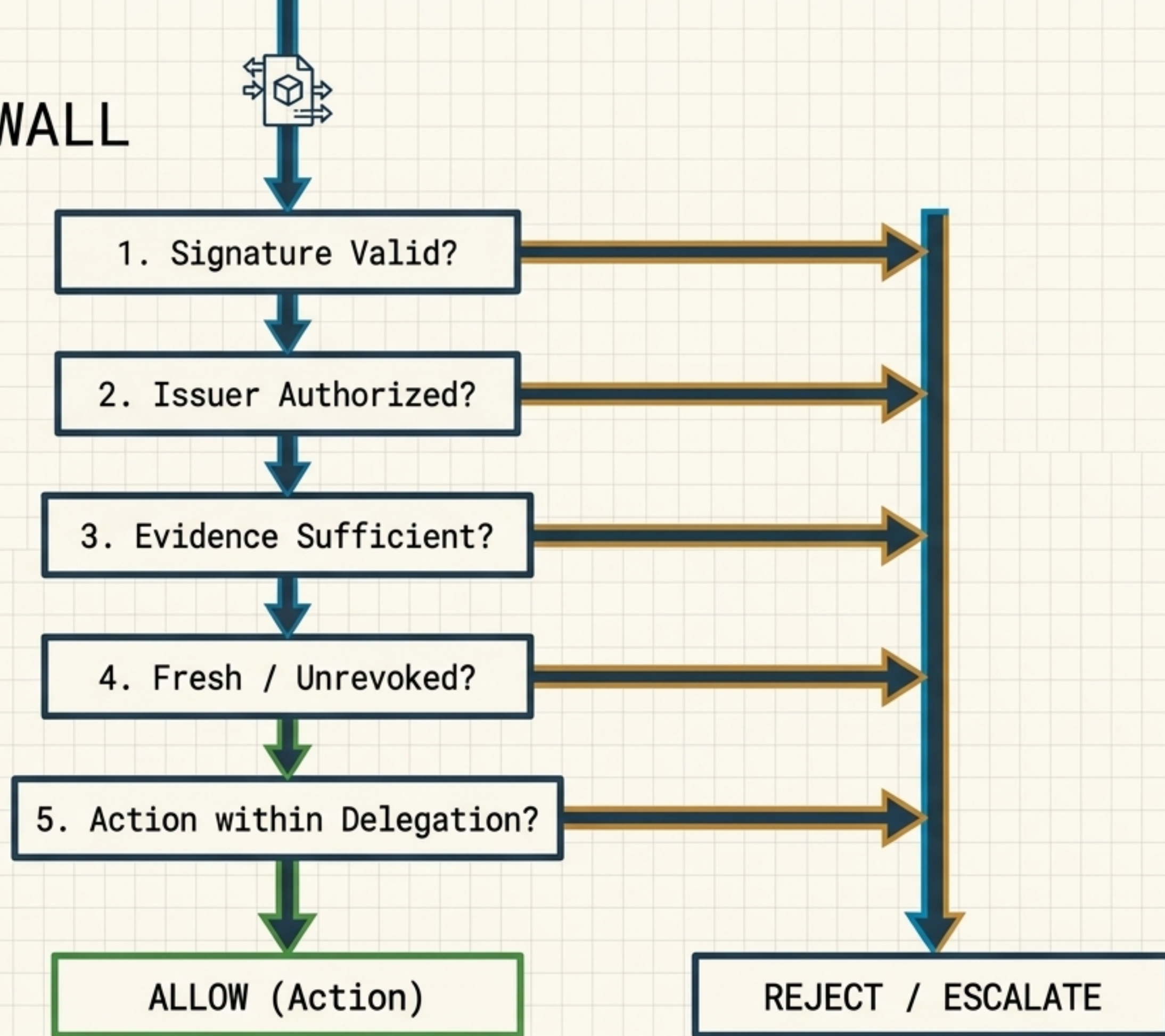
Failure State



Signature valid, **Authority false**

THE CONTEXT FIREWALL

Never trust remote AI because it is intelligent. Trust only the verifiable context.



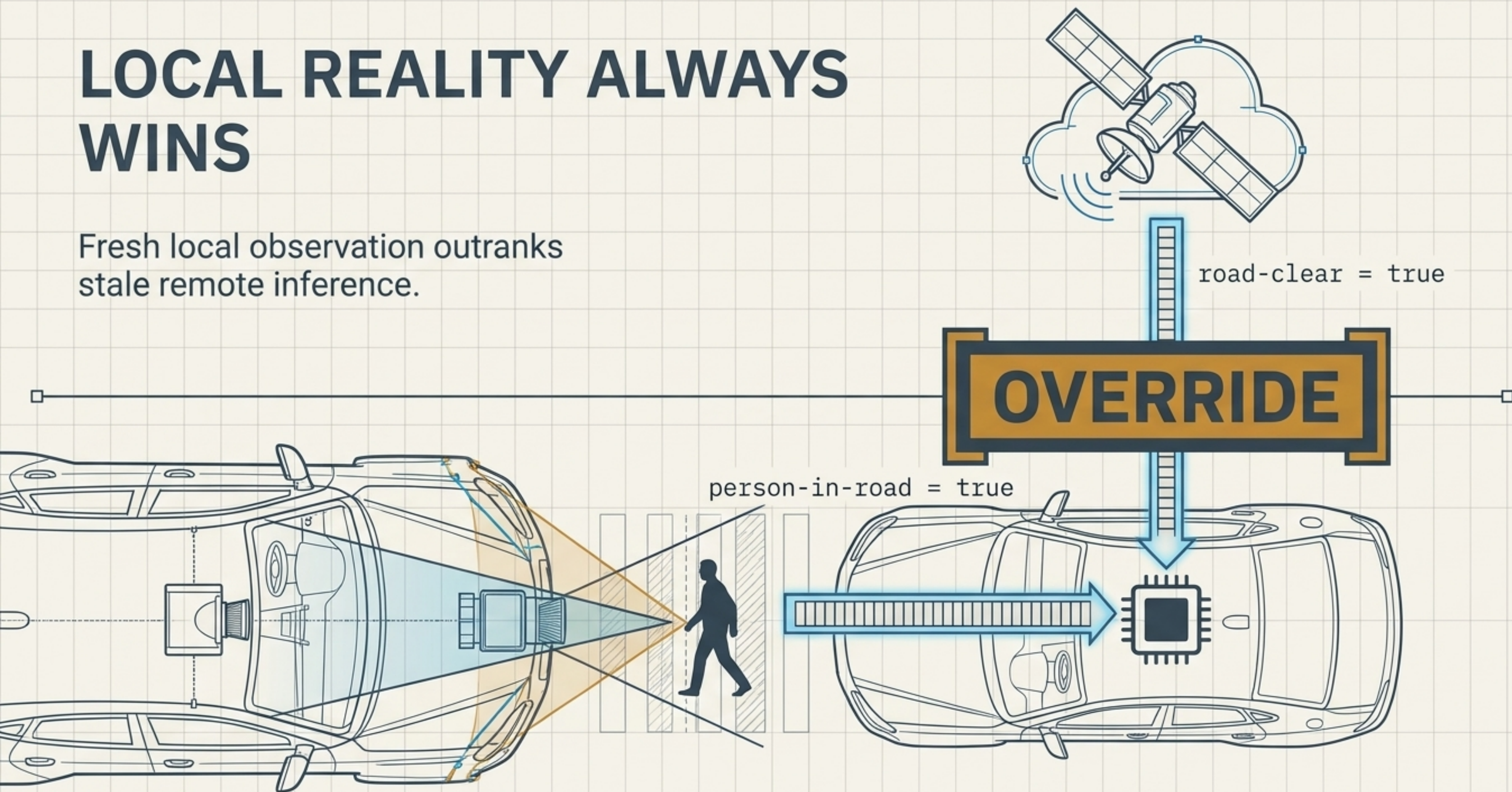
TRUST POLICIES REMAIN SOVEREIGN AND LOCAL

THE TRUST SPECTRUM

Local Node	Tolerance	Required Evidence	Acceptable Packet Types
Hospital-A	Zero	2 Accredited Sources	Highly Constrained Decision Packets
Research Agent	High	Unverified Hypotheses	Exploratory Generative Packets
Industrial Controller	Low	Manufacturer Signatures	Rigid Rule Packets

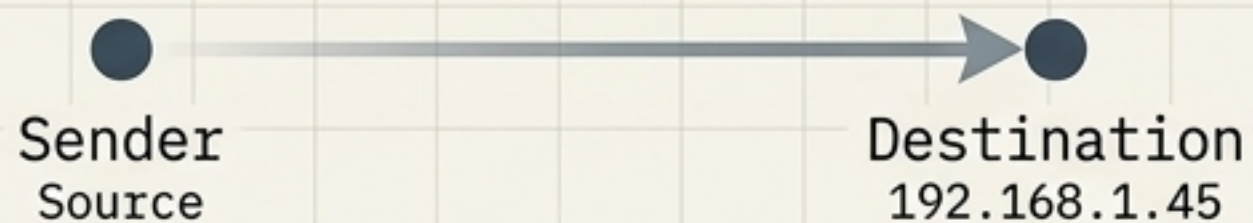
LOCAL REALITY ALWAYS WINS

Fresh local observation outranks stale remote inference.



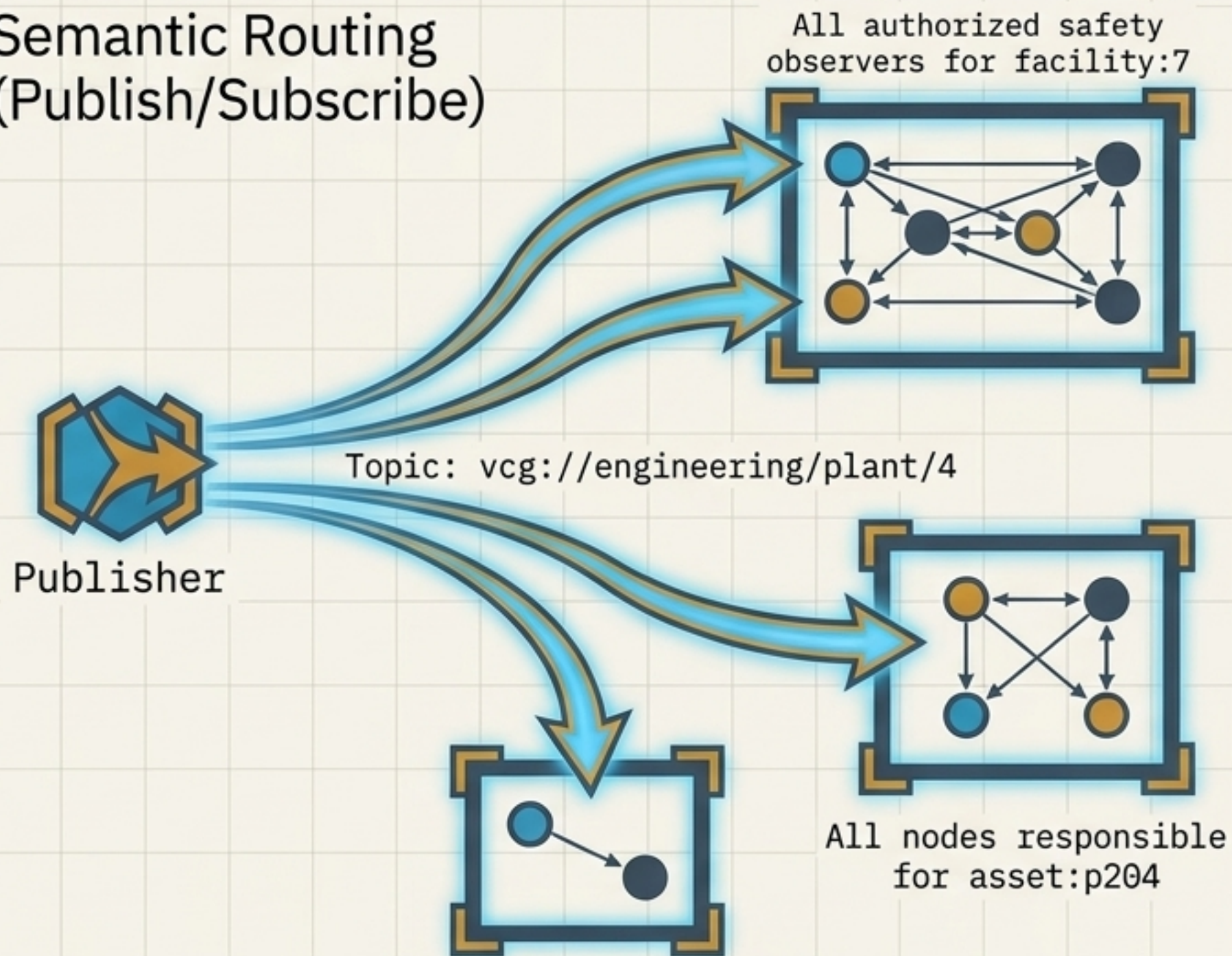
SEMANTIC ROUTING: ADDRESSING BY MEANING

Traditional Routing (Push)



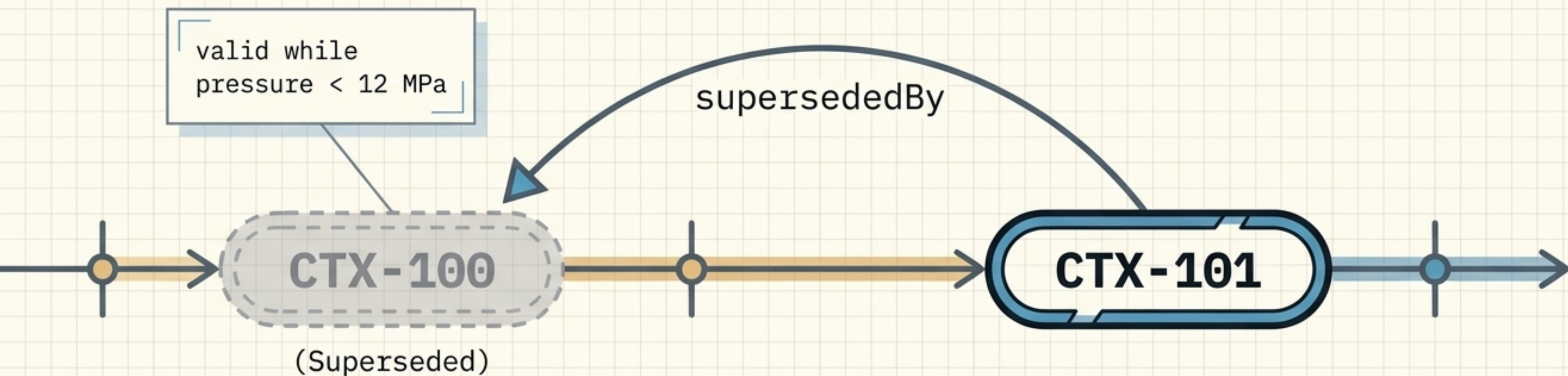
Send to IP: 192.168.1.45

Semantic Routing (Publish/Subscribe)



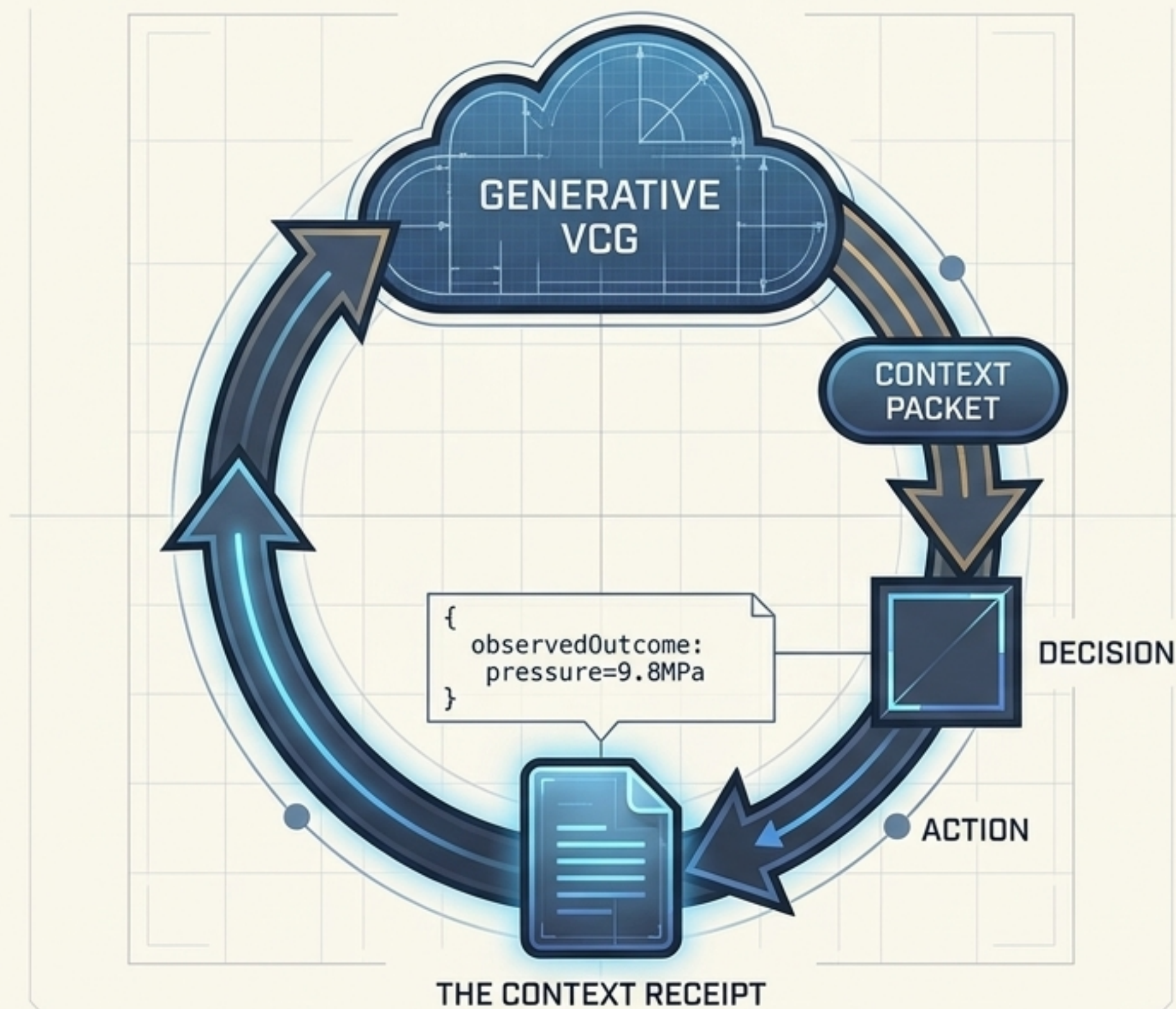
VALIDITY, REVOCATION, AND SUPERSESSION

Context creates a clear decision history for perfect audibility and scientific reproducibility.



CLOSING THE LOOP: THE CONTEXT RECEIPT

The graph learns from reality.



HUMAN-MACHINE SHARED CONTEXT

MACHINE VIEW

```
{  
  "purpose": "stop-pump",  
  "evidence": "sensor-readings",  
  "authority": "policy-4.2"  
}
```

HUMAN VIEW

RECOMMENDED ACTION:

Stop Pump P-204

Why: Vibration threshold exceeded

Authority: Plant Safety Policy 4.2

STOP PUMP

THE INTERNET OF UNDERSTANDING

THE CURRENT AI WEB

- Exchanges Data/JSON/Prompts
- Implicit Trust
- Ambiguous Authority
- Prompt -> Model -> Answer
- Global Access & Direct Control

THE INTERNET OF INTELLIGENCE

- ✓ Exchanges Meaning & Context Packets
- ✓ Verifiable Evidence
- ✓ Explicit Capability & Delegation
- ✓ Reality -> VCG -> Packet -> Decision -> Action
- ✓ Federated Sovereignty

THE CORE PROPOSITION

Generate globally. Verify structurally. Package minimally.
Distribute sovereignly. Decide locally. Return outcomes.

